

Report
Of
Java Full Stack Developer Internship

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Roll Number of Student: CS-2341826
Name of Student: AYUSH
Batch: 2026-27

CANDIDATE'S DECLARATION

I, Ayush Verma do hereby declare that the internship report titled “Java Full Stack Developer Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature_____

Student Name: Ayush

Roll No.: CS-2341826

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature: _____

Student Name: Ayush Verma

Roll No.: CS-2341826

INDEX

Chapter 1: Candidate's Declaration	ii
Chapter 2: Acknowledgment	iii
Chapter 3: Internship Completion Certificate	iv
Chapter 4: Project Description	1
4.1 Introduction	
4.2 Organization Profile	
4.3 Problem Statement	
4.4 Project Objectives	
4.5 Scope of the Project	
Use of 4.6 Technologies and Tools Used	
4.7 System Architecture	
4.8 Methodology (Detailed Weekly Breakdown)	
4.9 Expected Outcomes	
4.10 Certificates of completion	
Chapter 5: Bibliography/References	12

Chapter 4: Project Description

4.1 Introduction

This report outlines the internship conducted at EduSkills in Java Full Stack Development in an 8-week period in a virtual environment. From basic academic programming theory to industry-standard full-stack development is one of the most important steps in a computer science engineering curriculum that modern enterprise web applications require to deliver robust, scalable and highly secure architecture.

The decoupled, microservice-based architecture is the new trend in the software world, wherein the front end presentation layer and the back end business logic operate independently, with the former communicating with the latter through RESTful API. This internship programme is especially created to fill the gap between theory and these modern industry practices. It did not have a single client project, but rather gave an intensive learning experience, module-wise, covering the whole technological spectrum. It started with creating responsive front end user interfaces with HTML, CSS and Bootstrap, moved on to dynamic scripting with JavaScript and ended with developing secure backend enterprise applications using Spring Boot and integrating relational databases using Hibernate ORM.

4.2 Organization Profile

As a key AICTE supported organisation, EduSkills has a vision of 'Nation Building Through Skills'. It focuses on delivering structured virtual internships with hands-on experience on new technologies. EduSkills is an ISO 9001:2015 certified firm that works closely with schools throughout the country to significantly improve students' employability.

The organization has the ability to mimic industry needs and offer weekly evaluations, which will help students develop proven industry-related technical skills. Rather than just teaching syntax, EduSkills courses emphasize best practices in software development,

version control and system architecture, making the leap between the classroom and corporate expectations.

4.3 Problem Statement

The main issues that were discussed in this training were getting to grip with the end-to-end development process for an enterprise-level web application. One of the challenges that engineering students often face is to move from writing discrete scripts to writing integrated systems where various distinct technologies need to communicate seamlessly.

The internship was centered around the holistic development of skills, and the main "problem" faced was successfully merging all the different technical lines, namely HTML/CSS, JavaScript, Core/Advanced Java, Spring Boot, and MySQL, into a single skill set that could be used in creating, deploying, and maintaining full-stack software solutions. The difficulties ranged from learning about HTTP state management to ensuring that no credentials are leaked when connecting to the database, to structuring the relational data without redundancy, to writing scalable business logic that could serve concurrent requests from users.

4.4 Project Objectives

To design, develop and build responsive, mobile-first frontend user interfaces using HTML5, CSS3, Bootstrap and JavaScript.

To build solid knowledge of Core Java concepts including OOP, Collections framework, Exception handling and effective Multithreading.

To build enterprise level back end APIs using Advanced Java concepts, Servlets and powerful Spring Boot framework for fast application development.

- Implementing secure, optimized and persistent database operations using MySQL and Object-Relational Mapping (ORM) using Hibernate to abstract raw SQL from the database.

To implement and rigorously follow version control conventions for the codebase in the version control system (VCS) Git, for safe codebase management and tracking incremental software changes.

4.5 Scope of the Project

This virtual internship was to cover the full spectrum of the Java Full Stack ecosystem. The learning pathway was very complex. It was a client side UI design that was guaranteed cross-browsers, server side business logic implementation with adherence to the MVC design pattern, and backend database administration with schema design and relational mapping.

Further, knowledge of deployment, configuration, and debugging of these enterprise applications was added in integrated development environments (IDEs) such as VS Code and Eclipse. The scope was closely monitored by taking weekly grade point checks, which ensured the continuous level of proficiency and allowed for the successful implementation of concepts into actual code.

4.6 Technologies and Tools Used

The technology stack used for the internship is well-known in the industry as the de-facto architecture for enterprise applications written in Java. Every tool was chosen to perform its function in the three-tier architecture.

Frontend Technologies:

HTML5 & CSS3: The basic blocks of creating web structure and styling. Used semantic tags for accessibility and CSS modules for scalable styling.

Bootstrap: A front-end framework that allows the creation of responsive and mobile-first websites using a strong and comprehensive grid system and pre-made components.

JavaScript & jQuery: Client-side scripting languages used for DOM manipulation, client-side form processing and for fetching data asynchronously (AJAX) to provide the user with dynamic experiences.

Backend Frameworks:

Core Java (JDK17+): Used to develop the business logic, leveraged Collection API advanced data structures, and maintained thread safety.

Advanced Java (JDBC & Servlets): Gave students the background knowledge of how Java communicates with Web servers (such as Tomcat) and how it can use an external data source.

Spring Boot: Extension of the Spring framework with an aim of eliminating boilerplate configuration, embedding servers, and auto-configuration to quickly expose RESTful APIs.

Database & Tools:

MySQL: A powerful relational database management system (RDBMS) for securely storing data.

Hibernate (ORM): To map java objects directly to mysql database tables, which will reduce the need of manual sql query writing drastically.

Git: Distributed version control system for tracking changes to the source code during development.

4.7 System Architecture

The main focus of the architectural approach of the training modules was centred around the Model-View-Controller (MVC) paradigm and the classic three-tier system architecture. This way of doing things will make the application highly maintainable and scalable while keeping the concept of separation of concerns intact.

1- Presentation Layer (View / Frontend):

This tier is for all interactions with the user. It receives inputs through HTML form, validates them with JavaScript and dynamically displays the server's response. It has no interaction with the backend logic whatsoever, it only gets/sends data via http (GET/POST/PUT/DELETE) calls.

This layer is known as the Application Layer (controller & service / backend).

Developed with Spring Boot. The Controller receives the incoming HTTP request, matches it against a set of endpoints and sends the processing request to the Service layer. The Service layer has the core business logic, does calculations, and applies business rules prior to requesting data changes.

3. Data Access Layer (Model / Database):

This layer deals with how data is stored. Java objects (Entities) are mapped directly to MySQL tables by using Hibernate ORM and Spring Data JPA. The repository interfaces are responsible for CRUD (Create, Read, Update, Delete) transactions, including supporting atomic, consistent, isolated and durable (ACID) properties.

4.8 Methodology (Detailed Weekly Breakdown)

The internship had a very rigid methodology and lasted for 8 weeks. It progressed from basics of the front-end to the enterprise level of the back-end of the systems each week.

Week 1: HTML Fundamentals and Structure.

The first part of the program involved creating the backbone of web applications. The topics covered were semantic HTML5 elements, integration of multimedia elements, construction of forms and tabular data representation. Problems involved were creating usable and well-structured “static” web pages without styling.

Once the structure was set, attention was turned to visual responsiveness and aesthetics. Week 2 explored the Box Model, absolute and relative positioning, Flexbox and CSS Grid in detail. In week 3, Bootstrap was introduced, which significantly reduced the time to style.

The major learning took place with the 12-column mobile-first grid system implementation. Using the Bootstrap utility classes, complex responsive navbars, cards, and modal windows were built with little custom CSS.

Week 4: JavaScript & jQuery

This module was a transition module into dynamic scripting, client-side logic. We learned about variable scoping (let vs const), arrow functions, array methods and Object Oriented JavaScript.

Much of the week was spent learning how to do Document Object Manipulation (DOM) to capture user clicks, manipulate text dynamically, and stop default form submissions for client side validation. The use of jQuery was introduced to show how little boilerplate code can be used to traverse the DOM.

Week 5: Core Java Deep Dive. With the implementation of the backend, week 5 thoroughly exercised Object-Oriented paradigms. It went beyond mere syntax, and introduced concepts such as polymorphism, inheritance and how encapsulation is crucial in enterprise software.

They spent a considerable amount of time learning to use the Java Collections Framework (ArrayList, HashMap, HashSet) to store and retrieve data quickly and efficiently. The use of the exception handling (try-catch-finally) construct was highlighted to create fault-tolerant programs.

The following is an example of standard Java Encapsulation Practice. The following is an illustration of a standard Java Encapsulation Practice.

```
public class Employee {  
  
    private Long id;  
  
    private String name;  
  
    public Employee(Long id, String name) {  
  
        this.id = id;  
  
        this.name = name;  
  
    }  
}
```

```

public String getName() {

    return this.name;

}

}

```

Week 6: HOCKH Advanced Java (JDBC & Servlets)

Before using automated frameworks, it was important to know the basic hawking of the web servers and database links. Introduced Java Database Connectivity (JDBC) which uses DriverManager, Connection, and PreparedStatement objects in order to safely query databases without being vulnerable to SQL injection attacks.

The Servlet lifecycle (init, service, destroy) was discussed and how java application runs on a web container at low level (apache tomcat) was explored.

Week 7: Spring Framework and Spring Boot

The 7th week was the most focused and fulfilling week. Spring Boot has greatly reduced the amount of boilerplate code that was previously required for actual Servlets.

Understood concepts of Inversion of Control (IoC) and Dependency Injection (DI), and applied the spring container to control the lifecycle of objects. Built RESTful APIs with standard Annotations.

This is an example of a Spring Boot REST Controller. The following is an example of a Spring Boot REST Controller.

```

@RestController

@RequestMapping("/api/users")

public class UserController {

```

```

@Autowired

private UserService userService;

@GetMapping("/{id}")

public ResponseEntity<User> getUserById(@PathVariable Long id) {

    return ResponseEntity.ok(userService.findById(id));

}

}

```

In week 8, students will learn how to use Hibernate ORM and MySQL, while also using version control.

At the end of the week the architecture was consolidated. When you take Hibernate as an ORM, it's a complete change of the database interaction paradigm. JPA annotations such as `@Entity` and `@Table` were used to map Java objects (Entities) to the tables in the database, rather than writing SQL strings.

The relational database used was MySQL. Finally, there were strict version control procedures with Git (commit, push, branch, merge) to set up for real world collaborative development.

4.9 Expected Outcomes & Key Learnings

Upon finishing the program, the participant was able to demonstrate a practical proficiency in the full range of Java Full Stack skills. The immediate results were:

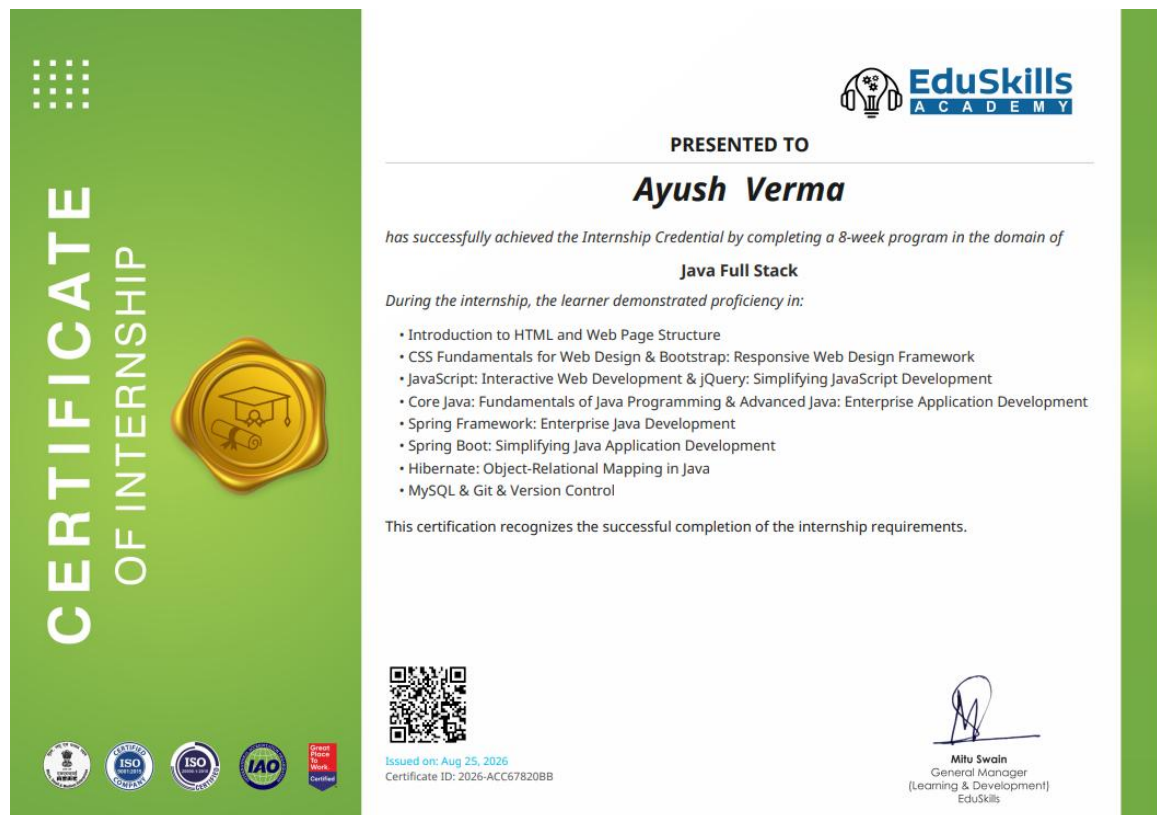
Technical skills: Ability to design independent web apps from scratch, determining folder structure, routing decisions, and database schema.

2. Deep knowledge of how data moves from a user's browser, through a Java-based REST API down to a relational database and back.

3. Knowledge of how to effectively troubleshoot and solve complex stack traces in Spring Boot and dependency issues with build tools, such as Maven.

5. Passing of rigorous weekly grade point assessments culminating in the awarding of the recognized Virtual Internship Certificate from EduSkills.

4.10 Certificates of Completion



Chapter 5: Bibliography/References

1. EduSkills AICTE Virtual Internship Program Documentation & Weekly Training Modules.
2. Official Java SE Documentation (Oracle) for Core and Advanced Java Application Programming Interfaces (APIs). [Online]. Available: <https://docs.oracle.com/en/java/>
3. Spring Framework Official Reference Documentation (Spring.io) for dependency injection and REST API development. [Online]. Available: <https://spring.io/projects/spring-boot>
4. Hibernate ORM Official Reference Guide for object-relational mapping syntax. [Online]. Available: <https://hibernate.org/orm/documentation/>
5. Bootstrap Official Documentation for responsive grid system implementations. [Online]. Available: <https://getbootstrap.com/docs/>